

Full Report: Refined 3D Index of m004

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1 Manifold Summary

- Manifold: m004
- Number of tetrahedra: $n = 2$
- Number of cusps: $r = 1$
- Independent easy edges: 0
- Hard edges (fugacity variables): 1
- q -order (half): 10 (q -order: 10/2)

2 SnaPy Gluing Equations

The gluing equation matrix has rows:

- Rows 0...1: edge equations
- Rows 2, 4, ...: meridian equations (per cusp)
- Rows 3, 5, ...: longitude equations (per cusp)

Columns are ordered as $Z_1, Z'_1, Z''_1, Z_2, Z'_2, Z''_2, \dots, Z_2, Z'_2, Z''_2$.

$$G_{\text{glue}} = \begin{pmatrix} 2 & 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 & 2 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -2 & 2 \end{pmatrix}$$

Cusp Equations

Cusp 0:

$$\begin{aligned} \mu_0 &= Z_1 - Z'_2 \\ \lambda_0 &= -2 Z'_2 + 2 Z''_2 \end{aligned}$$

3 Edge Classification

Total SnaPy edge equations: 2. Independent edge basis rank: $n - r = 1$. Of these, 0 are **easy** and 1 are **hard**.

3.1 Hard Edges

Hard edges contribute fugacity variables $\eta_0, \eta_1, \dots$ to the refined index.

$$1. \ C_{\text{hard},0} = 2 Z_1 + Z'_1 + Z_2 + 2 Z''_2$$

Basis ordering in g_{NZ} (position block rows): meridians (r rows), hard edges, easy edges.

4 Neumann–Zagier Data

4.1 Symplectic Matrix g_{NZ}

Row ordering: meridians $(\mu_0, \dots, \mu_{r-1})$, hard edges, easy edges | longitudes/2 $(\lambda_0/2, \dots)$, Γ vectors.
 Column ordering: $Z_1, \dots, Z_n, Z_1'', \dots, Z_n''$ (block form).

$$g_{NZ} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & -1 & 2 \\ 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

4.2 Inverse g_{NZ}^{-1}

$$g_{NZ}^{-1} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 2 & 0 & -1 & -2 \\ 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 1 \end{pmatrix}$$

4.3 Affine Shifts

$$\nu_x = (-1, -1)$$

$$\nu_p = (-1, 0)$$

5 Local Charges Formula

The refined index is computed by the formula:

$$I^{\text{ref}}(q; \eta_0, \dots, \eta_{k-1}) = \sum_{e_{\text{int}} \in (\frac{1}{2}\mathbb{Z})^{n-r}} \left[\prod_{a=0}^{k-1} \eta_a^{e_{r+a}} \right] \cdot (-q^{1/2})^\varphi \cdot \prod_{j=1}^n I_\Delta((g_{NZ}^{-1}\kappa)_j, (g_{NZ}^{-1}\kappa)_{n+j}) \quad (1)$$

where:

- $\kappa = (m_{\text{ext}}, \underbrace{0, \dots, 0}_{n-r}, e_{\text{ext}}, e_{\text{int}})$ is the $2n$ -vector of external + internal charges.
- The **local charges** (tetrahedron arguments) are $(m_j, e_j) = (g_{NZ}^{-1}\kappa)_j$ and $(g_{NZ}^{-1}\kappa)_{n+j}$.
- The **phase exponent** is $\varphi = m_{\text{full}} \cdot \nu_p - e_{\text{full}} \cdot \nu_x$.
- $I_\Delta(m, e)$ is the tetrahedron index.
- $k = \text{num_hard}$ is the number of hard edges carrying fugacity variables.

6 Non-Closable Cycles & Basis Selection

Cusp 0: non-closable cycles found: $(1, 0)$.

Selected Basis

- Cusp 0: slope $(1, 0) \Rightarrow m = 1, e = 0$ [default]

7 Weyl Symmetry & Dehn Filling Compatibility

Convention: $f(\eta) = \eta^{b \cdot m + a \cdot e} \cdot I(m, e)$, $f(\eta) = f(\eta^{-1})$.

$$a = \left(\frac{1}{2}\right), \quad b = \left(-\frac{1}{2}\right)$$

- $a \in \mathbb{Z}$: $\times$
- $2b \in \mathbb{Z}$: $\checkmark$

Dehn filling not directly compatible: $a \notin \mathbb{Z}$.

Weyl symmetry check: $\times$ (29/45 entries pass).

8 Refined Index

Evaluated at q -order (half) = 10, giving series up to q^5 .

Output in **Weyl-manifest form**: $f(m, e) = \eta^{b \cdot m + a \cdot e} \cdot I(m, e)$, where the displayed $I(m, e)$ satisfies $I(\eta) = I(\eta^{-1})$.

$$\begin{aligned}
I(-2, -2) &= q^3 + q^3 \eta^{2v_0} + q^4 + q^4 \eta^{2v_0} + 2q^5 \\
I(-2, -1) &= \eta^{-v_0} (q^3 \eta^{v_0} + q^4 \eta^{-v_0} + 2q^4 \eta^{v_0} + q^5 \eta^{-v_0} + 2q^5 \eta^{v_0}) \\
I(-2, 0) &= \eta^{-2v_0} (q^3 + 2q^4 + q^5 \eta^{-2v_0} + 3q^5 + q^5 \eta^{2v_0}) \\
I(-2, 1) &= \eta^{-3v_0} (q^3 \eta^{-v_0} + 2q^4 \eta^{-v_0} + q^4 \eta^{v_0} + 2q^5 \eta^{-v_0} + q^5 \eta^{v_0}) \\
I(-2, 2) &= \eta^{-4v_0} (q^3 \eta^{-2v_0} + q^3 + q^4 \eta^{-2v_0} + q^4 + 2q^5) \\
I(-1, -2) &= \eta^{v_0} \left(-q^{1/2} \eta^{v_0} + q^{5/2} \eta^{v_0} + q^{7/2} \eta^{-v_0} + 3q^{7/2} \eta^{v_0} + 2q^{9/2} \eta^{-v_0} + 5q^{9/2} \eta^{v_0} \right) \\
I(-1, -1) &= -q - q^2 + q^3 + q^3 \eta^{2v_0} + q^4 \eta^{-2v_0} + 4q^4 + 2q^4 \eta^{2v_0} + q^5 \eta^{-2v_0} + 8q^5 + 2q^5 \eta^{2v_0} \\
I(-1, 0) &= \eta^{-v_0} \left(-q^{3/2} \eta^{-v_0} - q^{3/2} \eta^{v_0} + 2q^{7/2} \eta^{-v_0} + 2q^{7/2} \eta^{v_0} + 5q^{9/2} \eta^{-v_0} + 5q^{9/2} \eta^{v_0} \right) \\
I(-1, 1) &= \eta^{-2v_0} (-q - q^2 + q^3 \eta^{-2v_0} + q^3 + 2q^4 \eta^{-2v_0} + 4q^4 + q^4 \eta^{2v_0} + 2q^5 \eta^{-2v_0} + 8q^5 + q^5 \eta^{2v_0}) \\
I(-1, 2) &= \eta^{-3v_0} \left(-q^{1/2} \eta^{-v_0} + q^{5/2} \eta^{-v_0} + 3q^{7/2} \eta^{-v_0} + q^{7/2} \eta^{v_0} + 5q^{9/2} \eta^{-v_0} + 2q^{9/2} \eta^{v_0} \right) \\
I(0, -2) &= \eta^{2v_0} (-q - q^2 + q^3 \eta^{-2v_0} + q^3 + q^3 \eta^{2v_0} + q^4 \eta^{-2v_0} + 4q^4 + q^4 \eta^{2v_0} + 2q^5 \eta^{-2v_0} + 8q^5 + 2q^5 \eta^{2v_0}) \\
I(0, -1) &= \eta^{v_0} (-q \eta^{-v_0} - q \eta^{v_0} - q^2 \eta^{-v_0} - q^2 \eta^{v_0} + q^3 \eta^{-v_0} + q^3 \eta^{v_0} + 4q^4 \eta^{-v_0} + 4q^4 \eta^{v_0} + 8q^5 \eta^{-v_0} + 8q^5 \eta^{v_0}) \\
I(0, 0) &= 1 - 2q - 3q^2 + q^3 \eta^{-2v_0} + q^3 \eta^{2v_0} + 2q^4 \eta^{-2v_0} + 4q^4 + 2q^4 \eta^{2v_0} + 3q^5 \eta^{-2v_0} + 12q^5 + 3q^5 \eta^{2v_0} \\
I(0, 1) &= \eta^{-v_0} (-q \eta^{-v_0} - q \eta^{v_0} - q^2 \eta^{-v_0} - q^2 \eta^{v_0} + q^3 \eta^{-v_0} + q^3 \eta^{v_0} + 4q^4 \eta^{-v_0} + 4q^4 \eta^{v_0} + 8q^5 \eta^{-v_0} + 8q^5 \eta^{v_0}) \\
I(0, 2) &= \eta^{-2v_0} (-q - q^2 + q^3 \eta^{-2v_0} + q^3 + q^3 \eta^{2v_0} + q^4 \eta^{-2v_0} + 4q^4 + q^4 \eta^{2v_0} + 2q^5 \eta^{-2v_0} + 8q^5 + 2q^5 \eta^{2v_0}) \\
I(+1, -2) &= \eta^{3v_0} \left(-q^{1/2} \eta^{-v_0} + q^{5/2} \eta^{-v_0} + 3q^{7/2} \eta^{-v_0} + q^{7/2} \eta^{v_0} + 5q^{9/2} \eta^{-v_0} + 2q^{9/2} \eta^{v_0} \right) \\
I(+1, -1) &= \eta^{2v_0} (-q - q^2 + q^3 \eta^{-2v_0} + q^3 + 2q^4 \eta^{-2v_0} + 4q^4 + q^4 \eta^{2v_0} + 2q^5 \eta^{-2v_0} + 8q^5 + q^5 \eta^{2v_0}) \\
I(+1, 0) &= \eta^{v_0} \left(-q^{3/2} \eta^{-v_0} - q^{3/2} \eta^{v_0} + 2q^{7/2} \eta^{-v_0} + 2q^{7/2} \eta^{v_0} + 5q^{9/2} \eta^{-v_0} + 5q^{9/2} \eta^{v_0} \right) \\
I(+1, 1) &= -q - q^2 + q^3 + q^3 \eta^{2v_0} + q^4 \eta^{-2v_0} + 4q^4 + 2q^4 \eta^{2v_0} + q^5 \eta^{-2v_0} + 8q^5 + 2q^5 \eta^{2v_0} \\
I(+1, 2) &= \eta^{-v_0} \left(-q^{1/2} \eta^{v_0} + q^{5/2} \eta^{v_0} + q^{7/2} \eta^{-v_0} + 3q^{7/2} \eta^{v_0} + 2q^{9/2} \eta^{-v_0} + 5q^{9/2} \eta^{v_0} \right) \\
I(+2, -2) &= \eta^{4v_0} (q^3 \eta^{-2v_0} + q^3 + q^4 \eta^{-2v_0} + q^4 + 2q^5) \\
I(+2, -1) &= \eta^{3v_0} (q^3 \eta^{-v_0} + 2q^4 \eta^{-v_0} + q^4 \eta^{v_0} + 2q^5 \eta^{-v_0} + q^5 \eta^{v_0}) \\
I(+2, 0) &= \eta^{2v_0} (q^3 + 2q^4 + q^5 \eta^{-2v_0} + 3q^5 + q^5 \eta^{2v_0}) \\
I(+2, 1) &= \eta^{v_0} (q^3 \eta^{v_0} + q^4 \eta^{-v_0} + 2q^4 \eta^{v_0} + q^5 \eta^{-v_0} + 2q^5 \eta^{v_0})
\end{aligned}$$

$$I(+2, 2) = q^3 + q^3 \eta^{2v_0} + q^4 + q^4 \eta^{2v_0} + 2q^5$$